

Customer Churn EDA Report

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Project: Customer Churn Exploratory Data Analysis

1. Introduction

Customer churn refers to the situation where customers stop using a company's products or services. Understanding customer churn is important because high churn can affect revenue, customer loyalty, and long-term business growth.

This project performs Exploratory Data Analysis (EDA) on a Telco Customer Churn dataset. The objective is to identify patterns associated with customer churn using numerical analysis, categorical analysis, statistical summaries, and visualizations.

2. Dataset Details

Item	Details
Dataset	Telco Customer Churn
Rows after cleaning	7035
Columns after cleaning	20
Target variable	Churn
Churned customers	1864
Non-churned customers	5171
Overall churn rate	26.50%

3. Data Cleaning Summary

The dataset was examined for missing values, duplicate records, incorrect data types, and unnecessary columns.

Missing numerical values in tenure, MonthlyCharges, and TotalCharges were handled using median imputation. Missing categorical values were handled using the mode.

The TotalCharges column was converted to a numerical data type. Eight duplicate records were identified and removed. The customerID column was removed because it is an identifier and does not provide useful information for EDA.

After cleaning, the dataset contains 7035 rows and 20 columns. There are 0 missing values and 0 duplicate records.

4. Exploratory Data Analysis

4.1 Churn Distribution

The analysis found 1864 churned customers and 5171 non-churned customers. The overall churn rate is **26.50%**.

The churn count plot and percentage chart show that the non-churned customer group is larger than the churned group. However, the churned group represents an important proportion of the customer base.

4.2 Contract Type Analysis

The contract analysis shows differences in churn across contract types. The **Month-to-month** contract group has the highest observed churn rate of **42.67%**.

This suggests that contract type is an important feature associated with customer retention.

4.3 Numerical Feature Analysis

Feature	Non-Churn Mean	Churn Mean
Tenure	34.50	22.50
Monthly Charges	63.41	73.62
Total Charges	2554.35	1535.76

The numerical analysis compares tenure, monthly charges, and total charges between churned and non-churned customers. Differences between the two groups provide useful information for understanding churn behavior.

4.4 Other EDA Analysis

Internet service and payment method were compared with churn using categorical count plots. These analyses showed that churn rates vary across customer groups.

Histograms were used to study the distribution of tenure and monthly charges. Boxplots were used to examine the spread of numerical variables and identify potential outliers.

A correlation heatmap was used to study relationships between numerical variables. A scatter plot was also used to visualize the relationship between tenure and total charges.

5. Key Findings

- The overall customer churn rate is 26.50%.
- Month-to-month contract customers have the highest observed contract-level churn rate (42.67%).
- Customer tenure differs between churned and non-churned customers.
- Monthly charges show differences between churned and non-churned customers.
- Internet service type is associated with different churn levels.
- Payment method also shows differences in churn across customer groups.
- Correlation analysis reveals relationships among numerical variables.
- Boxplot analysis was used to identify potential outliers.

6. Business Recommendations

- Encourage month-to-month customers to move to longer-term contracts through discounts and loyalty benefits.
- Provide additional onboarding and support for customers with shorter tenure.
- Review pricing plans and provide suitable packages for customers with higher monthly charges.
- Investigate services associated with higher churn and improve service quality and technical support.
- Make billing and payment processes easier and more convenient.
- Use customer segmentation to identify high-risk customers and provide targeted retention campaigns.

7. Conclusion

This exploratory data analysis examined customer churn using a Telco Customer Churn dataset. The analysis included data cleaning, missing-value handling, duplicate removal, numerical analysis, categorical analysis, outlier analysis, correlation analysis, and multiple visualizations.

The analysis identified an overall churn rate of 26.50% and observed differences in churn behavior across contract type, tenure, monthly charges, internet service, and payment methods.

These findings can help the company identify high-risk customer groups and develop targeted retention strategies. Improving customer experience, service quality, pricing, and early customer engagement may help reduce churn.